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A Project Synopsis on

Mat Setu: A Decentralized Voting System Using Blockchain

Bachelor of Engineering

In

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Submitted by:

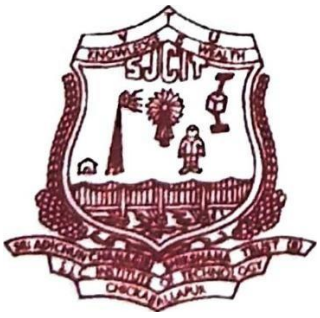
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ABSTRACT

Modern democratic frameworks depend on the electoral process as the foundational pillar of governance and institutional legitimacy. Traditional voting mechanisms, however, ranging from paper-based ballots to legacy Electronic Voting Machines (EVMs), continue to present significant operational and security vulnerabilities. Paper ballots remain susceptible to physical theft, stuffing, and logistical mismanagement, while legacy EVMs rely on centralized architectures that expose them to insider attacks, firmware manipulation, and persistent trust deficits among the electorate. A further challenge lies in enforcing "one person, one vote" without compromising voter privacy or enabling identity spoofing.

This project, titled MatSetu: A Decentralised Voting System, proposes a decentralized, multi-tier, blockchain-powered e-voting ecosystem built around two core technologies: an Ethereum-compatible decentralized ledger and Machine Learning-based biometric identity cross-referencing. The architecture decouples the identity layer from the ledger tier, preserving ballot secrecy while strictly preventing identity fraud. During registration, an on-premise face-api.js pipeline extracts a 128-dimensional facial descriptor for each voter and evaluates it against existing records using a Euclidean distance threshold ($d < 0.45$); a match within this variance aborts the registration, mitigating duplicate identities and Sybil attacks before they can reach the ledger.

Verified votes are cast as cryptographically signed, immutable transactions on a local, EVM-compatible Hardhat ledger governed by Solidity smart contracts, with a Proof of Authority (PoA) consensus model — using pre-approved validator nodes such as academic institutions or election commissions — replacing energy-intensive Proof of Work to deliver near-instant transaction finality without gas-fee volatility. Phase 1 of the project has focused on engineering and containerising this backend ecosystem: isolating services with Docker and Docker Compose, writing and compiling the Solidity election logic, configuring the Hardhat EVM environment, and implementing the biometric matching pipeline. The anticipated outcome is a tamper-resistant, Sybil-resistant voting platform that shifts electoral trust from centralized administrative oversight to mathematical consensus, directly furthering UN Sustainable Development Goal 16 on peace, justice, and strong institutions.

INTRODUCTION

In modern democratic frameworks, the electoral process serves as the foundational pillar of governance and institutional legitimacy. However, traditional voting mechanisms — ranging from paper-based ballots to legacy Electronic Voting Machines (EVMs) — consistently present significant operational and security vulnerabilities. Paper ballots are notoriously susceptible to physical theft, stuffing, and logistical mismanagement during transit and counting.

While legacy EVMs mitigate some physical tracking errors, they rely fundamentally on centralized architectures. This leaves them exposed to insider attacks, firmware manipulation, and severe trust deficits among the electorate. Furthermore, the modern digital landscape introduces a critical vulnerability: ensuring that "one person, one vote" is strictly enforced without compromising user privacy or enabling identity spoofing.

MatSetu bridges this socio-technical gap by introducing a decentralized, multi-tier, blockchain-powered e-voting ecosystem. The architecture fundamentally relies on two key technologies: Ethereum-compatible decentralized ledgers and Machine Learning-based biometric identity cross-referencing. The system decouples the identity layer from the ledger tier to maintain ballot secrecy while strictly preventing identity fraud. By utilising a local Hardhat node running an EVM-compatible execution environment, MatSetu provides an immutable, append-only repository for vote storage. Decentralization shifts the security paradigm from perimeter defence to mathematical consensus, entirely removing the "single point of failure" characteristic of centralized SQL/NoSQL electoral databases.

To secure the user entry point, the system implements an on-premise pipeline using face-api.js. This model extracts a 128-dimensional floating-point vector representing distinct spatial landmarks of the human face. The vector is mathematically evaluated against existing records during registration using a strict Euclidean distance threshold; if a match is found within a predefined variance, registration is aborted, mitigating duplicate identity or Sybil attacks before voters can interact with the ledger. Phase 1 of this project focuses specifically on the complete engineering and containerised orchestration of this backend ecosystem, while Phase 2 (outlined later in this synopsis) will extend the system with a voter-facing interface and public testnet deployment.

LITERATURE REVIEW

Author, Year	Title of Research Paper	Gap / Limitation in the Paper	How This Project Addresses the Gap
IEEE, 2024	*"Decentralized E-Voting Using Blockchain With Biometrics"*	Tested only in small-scale settings; biometric threshold not rigorously validated.	Defines a concrete distance threshold ($d < 0.45$) and containerizes services for independent scaling.
Minde et al., 2023 (IJCRT)	*"Vote Chain: Online Voting System Using Blockchain"*	Relies on conventional ID/OTP authentication, leaving identity spoofing unresolved.	Adds a biometric tier ahead of the ledger to block duplicate identities before wallet interaction.
Hajian et al., 2023 (MDPI)	*"Blockchain-Based E-Voting Systems: A Technology Review"*	A review paper; no working implementation or deployment challenges addressed.	Delivers a working, containerized end-to-end system rather than a conceptual survey.
Springer, 2025	*"Enhancing Security and Transparency in Online Voting through Blockchain"*	Limited focus on consensus cost/latency; assumes public PoW/PoS networks.	Uses a private PoA consensus among validators, removing gas-fee volatility and PoW costs.
Google Scholar, 2024	*"Blockchain Voting in Developing Democracies"*	Focuses on adoption/infrastructure; no specific technical or anti-fraud architecture.	Offers a lightweight, containerized architecture deployable on local infrastructure.

OBJECTIVES

The primary objective of this project is to design and develop MatSetu, a decentralised voting system that enforces "one person, one vote" while preserving ballot secrecy. The specific objectives are outlined as follows:

1. Decentralised, Tamper-Proof Vote Storage: Implement an Ethereum-compatible, EVM-based ledger — using Hardhat and Solidity smart contracts — that records every cast vote as an immutable, append-only transaction, removing the single point of failure inherent in centralized SQL/NoSQL electoral databases.

2. Biometric Identity Verification and Deduplication: Build an on-premise face-api.js pipeline that extracts 128-dimensional facial descriptors and applies a Euclidean-distance threshold ($d < 0.45$) to detect and reject duplicate voter registrations, enforcing "one person, one vote" and preventing Sybil attacks.

3. Secure, Non-Repudiable Ballot Casting: Enable voters to cryptographically sign ballots using ECDSA through a browser-based wallet (e.g., MetaMask), linking each vote to a one-time authorisation token so that votes remain authentic and cannot be modified or repudiated after submission.

4. Efficient, Low-Latency Consensus: Implement a Proof of Authority (PoA) consensus model using pre-approved validator nodes (e.g., academic institutions or election commissions) to remove gas-fee volatility and Proof-of-Work energy costs while achieving near-instant transaction finality.

5. Modular, Containerised System Architecture: Engineer the platform as isolated Docker microservices — a React client, a Node.js/Express server, a MongoDB instance, and a Hardhat EVM node — orchestrated via Docker Compose, ensuring consistent deployment and cross-container service discovery across development, testing, and staging environments.

By achieving these objectives, the project aims to deliver a mathematically verifiable, Sybil-resistant voting platform that reduces reliance on centralized administrative trust and strengthens confidence in institutional and academic electoral processes.

SYSTEM REQUIREMENTS

The implementation of **MatSetu** requires a combination of hardware, software, and network resources, structured to support containerised microservices, blockchain execution, and biometric processing.

Hardware Requirements

- A host machine capable of running Docker Engine v26.0 with sufficient RAM/CPU to run concurrent containers (React client, Node.js server, MongoDB, Hardhat EVM node).
- A webcam-equipped client device (desktop, laptop, or smartphone) for capturing the voter's facial image during registration and authentication.
- Persistent storage via Docker volumes for retaining MongoDB records across container lifecycles.

Software Requirements

- **Runtime Engine**: Node.js v22.x (LTS), natively handling asynchronous vector calculations.
- **Ledger Sandbox:** Hardhat v2.22 — a local EVM blockchain compilation toolchain with automated gas estimation and contract console tracing.
- **Smart Contract Language:** Solidity ^0.8.20, used to implement Election.sol (Candidate and Voter contract logic).
- **Biometric Framework:** face-api.js sitting atop a local TensorFlow (@tensorflow/tfjs-node) native binary bridge for rapid feature-vector generation.
- **Database:** MongoDB (document-oriented), storing 128-D facial descriptors, voting metadata, and election records.
- **Backend Framework:** Express.js REST API layer running on the Node.js server.
- **Client-Side Wallet Integration (Phase 2):** ethers.js or viem, together with the MetaMask browser extension, for ECDSA transaction signing.
- **Frontend (Phase 2):** React, for the voter-facing web interface.
- **Testing Tools (Phase 2):** Cypress and Selenium for automated end-to-end voter-journey testing.
- **Container Suite:** Docker Engine v26.0 with Docker Compose v2.x for isolated, reproducible application runtimes.

Network Requirements

- An internal **Docker** bridge network (voting_net) with embedded DNS-based service discovery for container-to-container communication (Node.js ↔ MongoDB ↔ Hardhat).
- Connectivity to the public **Ethereum Sepolia testnet** for the Phase 2 deployment and real-world latency/gas-fee testing.

METHODOLOGY

The methodology for developing the proposed MatSetu Decentralised Voting System involves multiple structured phases:

1. **Problem Definition and Requirement Analysis:**
   Clearly define the scope of the system, including the type of elections to be supported, target users (voters/administrators), and identification of core requirements such as ballot secrecy, identity verification, and vote immutability.

2. **System Architecture Design:**
   Design the overall multi-tier architecture, separating the authentication layer, the application/backend layer, and the blockchain ledger layer to ensure modularity and security.

3. **Environment Setup and Containerization:**
   Set up isolated development environments using Docker and Docker Compose, configuring separate containers for the database, backend server, and blockchain node to ensure consistent deployment.

4. **Biometric Pipeline Development:**
   Implement a facial recognition pipeline capable of extracting unique facial descriptors from voter images and matching them against existing records to prevent duplicate registrations.

5. **Smart Contract Development:**
   Design and develop the blockchain-based voting logic using Solidity smart contracts, defining rules for candidate management, vote casting, and vote counting.

6. **Blockchain Ledger Configuration:**
   Configure a local EVM-compatible blockchain environment and implement a consensus mechanism suited for institutional-scale voting, ensuring fast and reliable transaction processing.

7. **Backend and Database Integration:**
   Integrate the backend server with the database and blockchain layer to handle voter registration, authentication requests, and vote transaction processing in real time.

8. **System Integration:**
   Combine all components, including the biometric module, smart contracts, and database, into a unified system capable of handling the complete voter workflow.

9. **User Interface Development:**
   Design an intuitive interface that allows voters to register, verify their identity, and cast their vote easily, while providing administrators with election management tools.

10. **Testing and Deployment:**
    Perform rigorous testing in controlled environments to validate system performance, security, and reliability before deploying the system on a public test network for real-world evaluation.

EXPECTED OUTCOME

The expected outcome of this project is the development of a fully functional Decentralised Voting System that enhances electoral security, transparency, and accessibility in institutional governance. The key outcomes include:

- **Tamper-Proof Vote Storage:**
  The system will provide an immutable, append-only record of every cast vote by combining blockchain smart contracts with a locally hosted EVM ledger.

- **Fraud-Resistant Voter Verification:**
  The 128-dimensional biometric pipeline ensures that every registration is checked against existing records, reducing the chances of duplicate identities or Sybil attacks.

- **Faster, Low-Cost Consensus:**
  Election administrators and validators will receive near-instant transaction finality, helping votes be confirmed and tallied more quickly without volatile gas fees.

- **Reduced Administrative Errors:**
  By minimizing reliance on centralized databases and manual oversight, the system will help lower the risk of insider manipulation and human error in vote counting.

Overall, the system is expected to transform electoral processes by making them more secure, verifiable, accessible, and efficient, ultimately strengthening institutional trust and supporting modern, accountable governance systems.

APPLICATIONS

The proposed MatSetu platform has applications across several institutional and organisational settings:

- **Academic Institution Elections:**
  Student council, department, or hostel elections, where the institution itself can operate as a Proof-of-Authority validator node.

- **Government and Local Body Elections:**
  Municipal or cooperative-society polls supervised by an election commission acting as a trusted validating authority.

- **Corporate and Organisational Governance:**
  Shareholder voting, board elections, or union ballots that require an auditable, tamper-resistant record.

- **Referendum and Policy Polling:**
  Any bounded-participant decision-making process that needs "one person, one vote" guarantees with public verifiability but private ballot content.

These applications directly align with United Nations Sustainable Development Goal 16: Peace, Justice, and Strong Institutions, and specifically Target 16.6 (develop effective, accountable, and transparent institutions at all levels). By replacing administrative privileges with transparent, mathematics-based consensus, MatSetu addresses three impending threats to institutional trust — database tampering (countered by an append-only cryptographic ledger), ballot stuffing (countered by the 128-D biometric gatekeeper), and centralised points of failure (countered by distributed peer validation) — helping to build public confidence in digital governance systems.

CONCLUSION

The persistent vulnerabilities of traditional voting mechanisms, rising public distrust in centralized electoral systems, and the growing sophistication of identity fraud demand more secure and intelligent solutions in electoral governance. Conventional methods, while essential, are often constrained by centralized points of failure, susceptibility to tampering, and an inability to verify voter identity without compromising privacy. This project proposes MatSetu, a Decentralised Voting System, to address these challenges by combining the power of blockchain technology with real-time biometric identity verification.

By integrating an Ethereum-compatible decentralized ledger, Solidity smart contracts, and a deep-learning facial recognition pipeline, the proposed system delivers tamper-proof, transparent, and Sybil-resistant voting support. Unlike conventional electoral databases, the use of blockchain ensures that every cast vote is grounded in an immutable, cryptographically verifiable record, significantly reducing the risk of manipulation or fraudulent outcomes. The system not only safeguards the integrity of the ledger but also secures the voter entry point itself, enforcing "one person, one vote" through 128-dimensional biometric matching before a single vote is cast.

The expected outcomes, including tamper-resistant vote storage, elimination of duplicate registrations, near-instant transaction finality, and reduced administrative overhead, highlight the practical value of this system. Its applications across academic institutions, government bodies, corporate governance, and public referenda further demonstrate its broad impact and relevance.

In conclusion, this project goes beyond being just another blockchain application—it directly tackles a real-world problem where trust is the currency and mistakes are costly. If implemented properly, it can improve how elections are conducted, not by replacing institutional authority, but by making it more accountable and verifiable. It bridges the gap between cutting-edge decentralized technology and practical governance needs, contributing to a more transparent, secure, and accessible electoral system.

REFERENCE

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," decentralized manuscript, 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf

[2] M. Crosby, P. Pattanayak, S. M力を, and V. Kalyanaraman, "Blockchain Technology: Beyond Bitcoin," Applied Innovation Review, no. 2, pp. 6–19, Jun. 2016.

[3] N. Kshetri and J. Voas, "Blockchain-Enabled E-Voting," IEEE Software, vol. 35, no. 4, pp. 95–99, Jul./Aug. 2018. doi: 10.1109/MS.2018.2801546.

[4] R. L. Rivest, "The ThreeBallot Voting System," MIT Computer Science and Artificial Intelligence Laboratory, Cambridge, MA, Tech. Rep. MIT-CSAIL-TR-2006-059, Oct. 2006.

[5] F. P. Hjalmarsson, G. K. Hreidarsson, M. Hamdaqa, and G. Hjálmtýsson, "Blockchain Based E-Voting System," in Proceedings of the 2018 IEEE International Conference on Cloud Computing (CLOUD), San Francisco, CA, 2018, pp. 983–986. doi: 10.1109/CLOUD.2018.00151.

[6] A. Benaloh, "Verifiable Secret Ballot Elections," Ph.D. dissertation, Department of Computer Science, Yale University, New Haven, CT, Tech. Rep. YALEU/DCS/TR-560, Sep. 1987.

[7] D. Chaum, "Untraceable Electronic Mail, Return Addresses, and Digital Pseudonyms," Communications of the ACM, vol. 24, no. 2, pp. 84–90, Feb. 1981. doi: 10.1145/358549.358563.

[8] G. Wood, "Ethereum: A Secure Decentralised Generalised Transaction Ledger," Ethereum Project Yellow Paper, open technical draft, 2014. [Online].
Available : https://ethereum.github.io/yellowpaper/paper.pdf

[9] J. Buchmann, "Digital Signatures," in Introduction to Cryptography, 2nd ed. Berlin, Germany: Springer-Verlag, 2004, ch. 13, pp. 201–224.

[10] T. Kohno, A. Stubblefield, A. D. Rubin, and D. S. Wallach, "Analysis of an Electronic Voting System," in Proceedings of the 2004 IEEE Symposium on Security and Privacy, Oakland, CA, 2004, pp. 27–40. doi: 10.1109/SECPRI.2004.1301209.